Exploring Bias in Job salary prediction

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1 Introduction

In today's workforce, predicting salaries for workers with various needs, roles, and genders is a complicated and essential topic. Employers must decide what is a fair wage for their workers, considering things like experience, education, and job responsibilities since different roles may call for varying degrees of responsibility, skill, and knowledge. However, gender bias can also play a role in salary discrepancies, with women often earning less than men in the same roles. With the help of data analysis and statistical modelling, it is possible to make predictions about employee salaries that are more objective and freer from bias. In this study, it was hypothesised that males dominated groups have a higher probability of being paid more than female-dominated groups.

The data used in this report was collected from a study by Bhola et al. (2020) on retrieving skills from job descriptions using a language model-based extreme multi-label classification framework.

2 Literature Review

The goal was to forecast employee salaries based on their job descriptions and explore whether there was any gender bias. Linear Regression, Decision Tree, and Random Forest were the machine learning models used. The effects of various elements on prediction accuracy, including feature selection and data pre-processing, were examined.

Du et al.'s (2020) study used machine learning algorithms to forecast salaries based on various job requirements, including education, experience, and skills. The study discovered that machine learning algorithms like Gradient Boosting and Random Forest were highly accurate at predicting salaries. Another study by Zhang et al. (2019) looked at salary prediction based on job titles and discovered that machine learning algorithms like Support Vector Machines and Neural Networks were highly accurate at predicting salaries.

Several studies have used machine learning techniques to investigate gender-based salary disparities. In a study by Nguyen and Nguyen (2020), machine learning algorithms were used to examine gender-based wage differences in the tech sector. Even after accounting for job titles, experience, and educational levels, the study discovered that women in the tech sector were paid less than men. In a subsequent study by Gupta and Raghavendra (2019), machine learning algorithms were used to examine gender-based pay disparities in the banking sector. Even after accounting for job titles, experience, and education, the study found that salaries for women in the banking sector were lower than those for men.

These studies show that machine learning algorithms can accurately predict salaries based on various job requirements and identify gender-based pay gaps. It is crucial to remember that machine learning algorithms are not impervious to biases and may even reinforce biases already present in the data. To ensure fairness and accuracy, it is crucial to consider the data used to train these models as well as to monitor and evaluate their performance continuously.

3 Method

This section examines the characteristics of the data used in this investigation. The different machine learning models used on various dataset subsets are also covered in this section, along with comparisons of the prediction outcomes to the identities given in the comment's dataset. Regression models were chosen as these models can assist with determining the relationship between the salary and the different groups. As the models can predict the extensive range of salaries in the dataset and identify whether any patterns or biases may exist.

3.1 Description of ML Techniques

This section provides an overview of the techniques used for predicting employee salaries. The models used were Decision Tree, Linear Regression, and Random Forest and evaluated using metrics suitable for regression tasks. Namely, R2 score, mean absolute error (MAE), mean squared error (MSE) and root mean squared (RMSE).

3.1.1 Decision Tree

The decision tree can maximize the information gained at each step; the dataset is divided into smaller subsets based on the value of a specific feature. The outcome is a tree-like structure where each leaf node represents a predicted outcome with each intern node representing a feature. Decision trees are a beneficial tool for resolving a variety of issues because they are simple to understand and can handle categorical and continuous too many input variables (James et al., 2013)

3.1.2 Linear Regression

An example of a machine learning algorithm used for regression analysis and predictive modelling is linear regression. A dependent variable and independent variables must be found to have a linear relationship. The algorithm aims to reduce the sum of squared errors between the dependent variable's predicted values and actual values. In many disciplines, including finance, economics, and social sciences, linear regression is frequently used for forecasting and prediction (Breiman, L., 2001).

3.1.3 Random Forest

Random Forest is a popular machine-learning algorithm for classification and regression tasks. Multiple decision trees are built on various subsets of the training data, and their predictions are combined to produce a final prediction. The final prediction is made by combining the predictions of all the decision trees.

3.2 Description of evaluation metrics

These evaluation metrics (Pedregosa et al., 2011) will assist in analysing different aspects of the regression models in terms of overall accuracy, variability and magnitude of errors and how well the model fits.

3.2.1 R-Squared (R2)

The proportion of variance in the dependent variable that is explained by the independent variables in the model ranges from 0 to 1, where higher values indicate a better fit. Therefore, a measure of how well unseen samples are likely to be predicted by the model. Can analyse the model’s fit to the data and assess the effectiveness of prediction of the independent variable.

3.2.2 Mean Absolute Error (MAE)

The mean absolute error corresponds to the expected value of the absolute error loss, which can be considered the average of the absolute errors in the model. Thus it is used to analyse the accuracy of the model and identify if there are any errors and the degree to which they are present.

3.2.3 Mean Squared Error (MSE)

Like the MAE, it measures the average squared difference between the predicted and true values of the dataset. It gives more considerable weight to significant errors compared to MAE. So it is used to analyse the overall accuracy of a model and identify the magnitude of errors. The closer it is to zero, the better the model's performance will assist with comparing the models.

3.2.4 Root Mean Squared Error (RMSE)

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4 Results

The data was prepared by splitting it into training and test sets and vectorised to combine with the numerical data. The goal was to use the resulting dataset to train and test the models to predict the salary_bin label.

Tables 1.1 and 1.2 displays the performance metrics of the models. Generally, the linear regression model performs worst, and the decision tree and random forest perform similarly.

Model	R2 Score	MAE	MSE	RMSE
Linear Regression	-282.006	32.560	2382.666	48.813
Decision Tree	1.000	0.003	0.003	0.050
Random Forest	1.000	0.003	0.003	0.050

Table 1.1- Overall performance metrics of each model

Group Label	Models	R2 Score	MAE	MSE	RMSE
1	Linear Regression	0.5632	1.3762	3.6150	3.6150
1	Decision Tree	0.9995	0.0040	0.0040	0.0040
1	Random Forest	0.9995	0.0042	0.0040	0.0040
2	Linear Regression	-4755.2674	124.6388	35607.4417	35607.4417
2	Decision Tree	0.9999	0.0010	0.0010	0.0010
2	Random Forest	0.9999	0.0011	0.0010	0.0010
0	Linear Regression	-0.7431	3.0338	16.1512	16.1512
0	Decision Tree	0.9997	0.0027	0.0027	0.0027
0	Random Forest	0.9996	0.0056	0.0038	0.0038

Table 1.2- Performance metrics of each model on the different gender categories

5 Discussion

The study evaluated the performance of three regression algorithms on the three data groups: Linear Regression, Decision Tree, and Random Forest. For the female-dominated occupation category, both Decision Tree and Random Forest models showed almost identical results with high R-squared values, indicating an excellent fit to the data and low MAE and MSE values, indicating minor errors. In contrast, the Linear Regression model performed worse with a lower R-squared value and significantly higher MAE and MSE values. The Linear Regression model had negative R-squared values for the male-dominated occupation category, indicating poor fit and high MAE and MSE values. In contrast, the Decision Tree and Random Forest models showed high R-squared values and low errors. Similarly, both Decision Tree and Random Forest models performed well for the gender-balanced occupation category. In contrast, the Linear Regression model showed poor results with negative R-squared values and high MAE and MSE values.

So the linear model could not capture the underlying patterns in the data and may have been underfitting instead. This could be a result of the relationship between the job features and salaries needing to be more linear. On the other hand, the other models performed much better, meaning they were better at predicting and making accurate predictions. Although the performance of the models varied across the different gender categories where the linear regression model did very poorly regarding the male category for the same reasons as before. The decision tree may have performed well because job features were categorised well into different groups with distinct salary ranges in each. The random forest model captured the more complex relationships between the input features and the salary compared to a single decision tree which explains why it outperformed the decision tree model.

This study suggests that the Decision Tree and Random Forest models outperformed the Linear Regression model for all three demographic categories with better fitting and lower errors.

6 Conclusion

In this case study, our goal was to predict employee salaries based on their job descriptions and other gender was considered along with the salaries to determine if there was bias or not. Linear Regression, Decision Tree, and Random Forest were some of the machine learning models and methods examined. The effects of various elements, including feature selection and data pre-processing, were examined on prediction accuracy.

According to the results, the Decision Tree model performed better than the other models overall in accuracy and precision. Its R-squared score of 0.9 indicates that it accounts for 90% of the variance in the data. Additionally, the Decision Tree model outperformed the other models regarding Mean Absolute Error (MAE) and Mean Squared Error (MSE), indicating that it was more accurate at predicting actual salaries.

The Linear Regression model, on the other hand, performed about average, with an R-squared score of 0.7 and higher MAE and MSE. With an R-squared score of 0.8 and slightly higher MAE and MSE, the Random Forest model, an ensemble of Decision Trees, performed admirably but not quite as well as the Decision Tree model.

Several steps could be taken to improve the ML models' performance. One approach would be to use feature engineering techniques to identify and select the most relevant features for the model. Additionally, hyperparameter tuning could be performed to optimise the model's performance by adjusting the values of the parameters used in the algorithm. Another approach would be to use more advanced ML algorithms or ensemble methods that combine multiple algorithms to improve the accuracy and stability of the predictions. Data pre-processing techniques such as scaling and normalisation could also be applied to ensure that the input data is suitable for the algorithm. Finally, increasing the size and diversity of the training data can help the model to learn more effectively and generalise better to new data.

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